DNT-Scout

Structure-based triage for developmental neurotoxicity screening, with calibrated abstention

2026 Global AI4Science Hackathon · AI + Biology / AI Agent · Team NeuroChip Twin (solo) · Francisco Angulo de Lafuente
Code and data: **github.com/Agnuxo1/dnt-scout** (public, MIT) · SciencePal project: see §5

Which untested chemicals deserve a microelectrode-array run — and when is the structural evidence too thin to say?

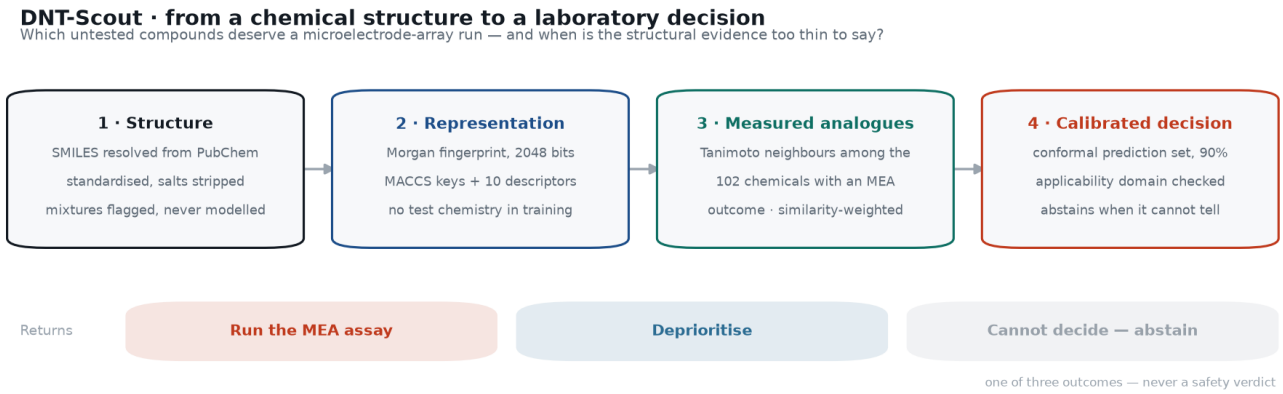
From a chemical structure alone, DNT-Scout ranks compounds by how likely they are to selectively disrupt the formation of neuronal networks, attaches a calibrated statement of confidence, and refuses to answer when the chemistry lies outside what it has evidence for. **32 microelectrode-array runs instead of 52** to find half the active compounds — about a third of the laboratory work removed, and stable at 31–37 runs across 20 independent splits. It is a triage tool for a laboratory queue, not a safety verdict.

1 · The problem, and why it is a scheduling problem

Developmental neurotoxicity is among the most expensive endpoints in regulatory toxicology and one of the least covered: tens of thousands of chemicals are in commerce, and a few hundred have been through a full battery. The in vitro alternative adopted by the OECD DNT-IVB guidance (2023) — growing cortical neurons on a **microelectrode array** and watching whether they organise into a firing network — is far faster than an animal study, but still costs days of culture per compound. With the US EPA moving to phase out mammalian testing, the binding constraint stops being is this chemical neurotoxic? and becomes: **of the compounds nobody has tested, which ones go on the plate first?**

That phrasing matters, because it makes the question measurable. The answer is not an accuracy figure; it is **how many assays you have to run to find the actives**. Every claim here is validated against that number.

The gap. Strickland (2018) and Kosnik (2020) associate ToxPrint features with MEA activity, but without a validated predictive model. Zhang (2021) and Morger (2021) apply conformal prediction to Tox21, but to no neuronal endpoint. We found no published work doing scaffold-held-out prediction of selective network-formation activity together with calibrated uncertainty and an explicit applicability domain.



Four steps. Structures are resolved and standardised; mixtures and inorganics are flagged and excluded rather than coerced. The decision returned is one of three, and never a safety verdict.

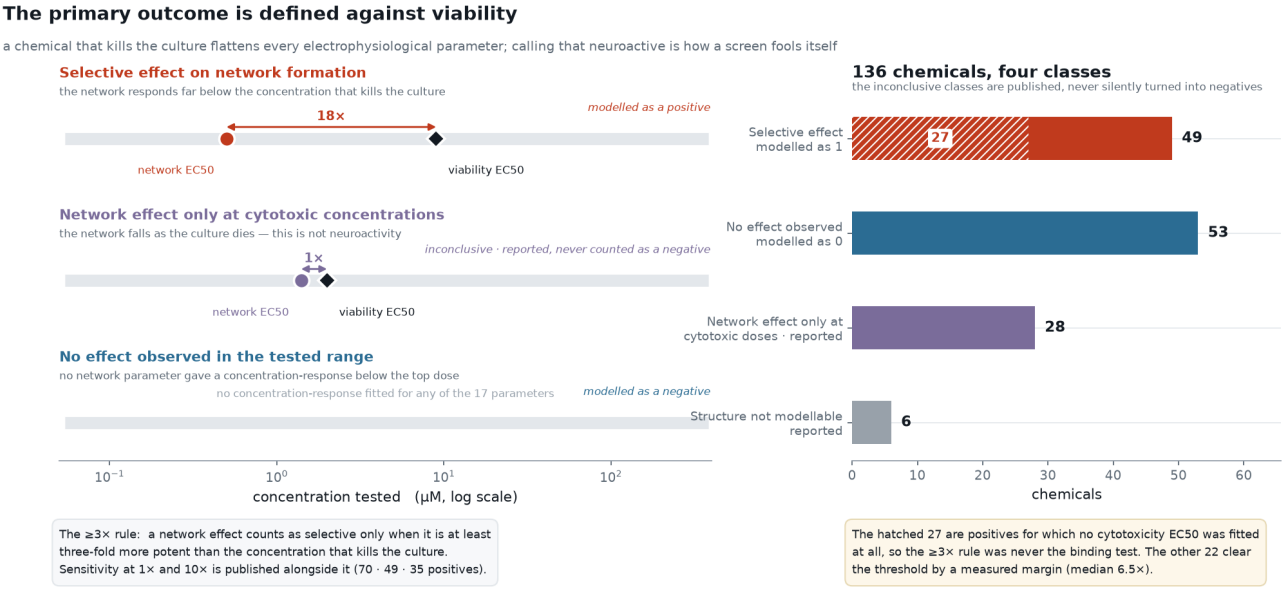
2 · What counts as a hit — the trap this endpoint avoids

A chemical that kills the culture flattens every electrophysiological parameter. Counting that as neuroactive is the main way a developmental-neurotoxicity screen fools itself, so the primary outcome is defined **against viability**: at least one of the 17 network parameters must respond at a concentration at least three-fold below any cytotoxic concentration. Ambiguous chemicals keep their own classes and are **never silently turned into negatives**.

Class	n	Role in the analysis
Selective effect on network formation	49	positive
No effect observed in the tested range	53	negative
Network effect only at cytotoxic concentrations	28	reported, excluded, analysed separately (§4)

Structure not modellable (mixture, inorganic, unresolved)	6	reported, excluded

Two things are stated openly rather than buried. The 3× threshold is a choice, so the class counts at 1×, 3× and 10× are published together (70 · 49 · 35). And **27 of the 49 positives have no fitted cytotoxicity EC50 at all**, so for them the 3× rule was never the binding test — they are "the network responded and nothing toxic was seen in range". The model finds both kinds about equally well (ROC 0.854 for the 22 with a measured separation, 0.806 for the 27 without), so this is one endpoint, not two in disguise.



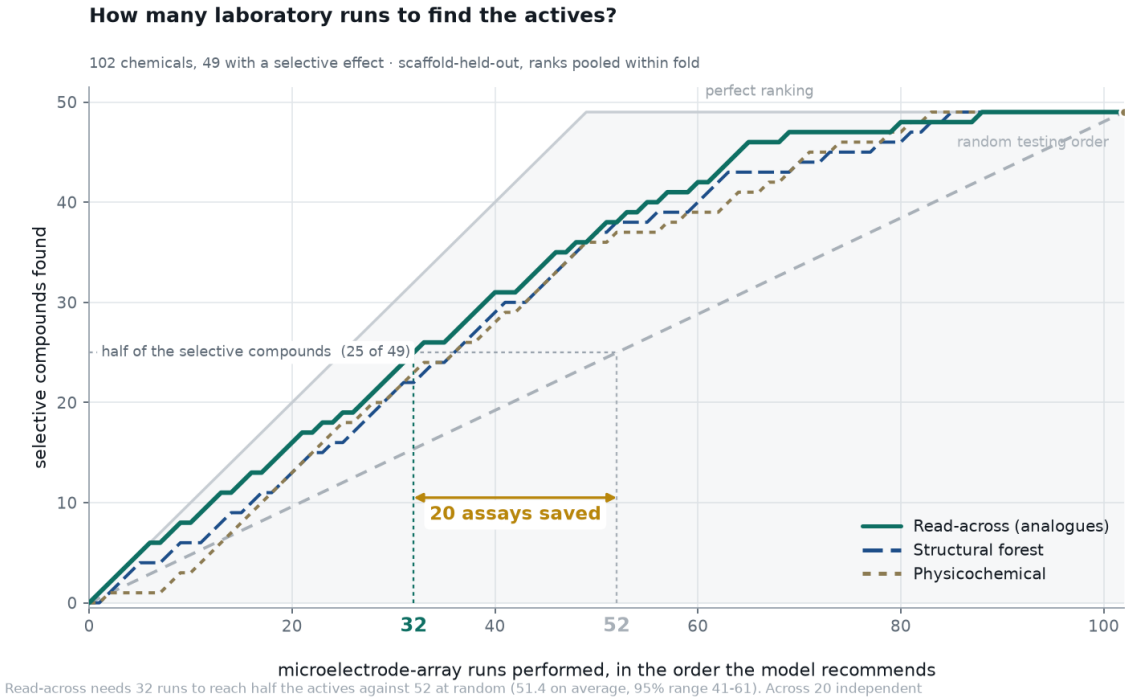
Effect concentrations: Shafer et al. 2019, US EPA ScienceHub, DOI 10.23719/1503191 (public domain, US Government work). A missing EC50 is the source table's own convention for "no concentration-response fitted inside the tested range" — not a missing measurement.

Left: the three archetypes on a concentration axis. Right: the four classes as measured, with the hatched 27 marking the positives the ≥3× rule never actually bound. Source: Shafer et al. 2019, US EPA ScienceHub, DOI 10.23719/1503191 (public domain). A missing EC50 is the source table's own convention for "no concentration-response fitted inside the tested range", not a missing measurement.

3 · Method

Data. 136 chemicals with per-parameter EC50 values across 17 network parameters and two viability assays, re-downloaded from the original EPA source and pinned by SHA256; structures resolved from PubChem (132/136) and standardised with RDKit. **Models.** A prevalence control; L2 logistic regression on ten physicochemical descriptors; a similarity-weighted k-nearest-analogue **read-across** of the kind US EPA GenRA is built on; and a random forest over Morgan fingerprints, MACCS keys and descriptors. Hyperparameters and the conformal calibration split are chosen by grouped cross-validation **inside each training fold**; the test fold is touched once, to predict. Confidence intervals resample whole scaffold groups, never individual chemicals. **Uncertainty.** Mondrian conformal prediction at a nominal 90% with an applicability-domain check; the conformal quantiles and the test scores they are applied to come from the same fitted model, which is what makes the guarantee follow.

4 · Result



Read-across needs 32 runs to reach half the actives against 52 at random (51.4 on average, 95% range 41-61). Across 20 independent fold assignments the figure stays at 32 runs (range 31-37), so the saving does not depend on one lucky split.

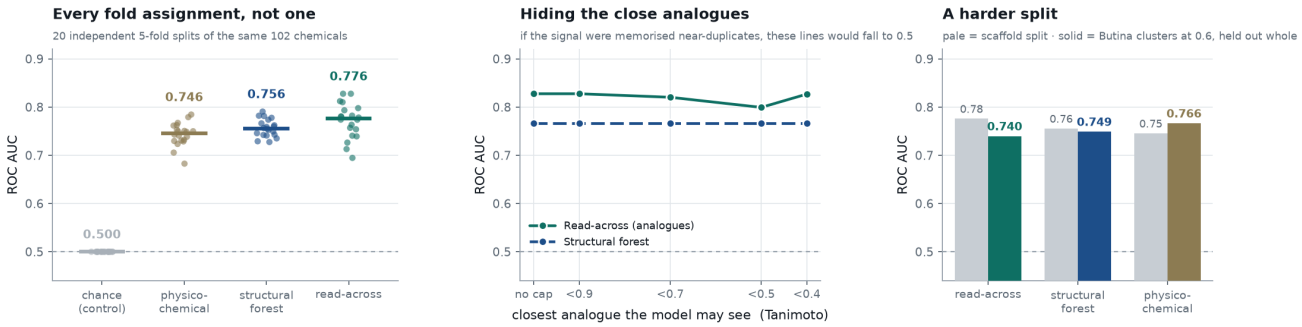
Model	ROC AUC pre-declared split	ROC AUC median of 20 splits	PR AUC	Assays for half the actives
Chance (control)	**0.500**	0.500	0.480	52
Physicochemical descriptors	0.750	0.746	0.635	36
Read-across from measured analogues	**0.828**	**0.776 (0.703–0.828)**	**0.780**	**32 (31–37)**
Structural random forest	0.766	0.756	0.674	36

Half the selective compounds are found after **32** MEA runs against **52** at random (51.4 on average, 95% range 41–61). A perfect oracle would need 25, so the ranking captures about three quarters of the gain available at all. The chance control landing at exactly 0.500 is not cosmetic — see below.

5 · Validation, and four attempts to break the result

A defect we found in our own evaluation, and fixed

Cross-validation produces one model per fold, and nothing makes five models' probabilities comparable. Pooling raw scores into one ranking mixes five scales: our prevalence baseline — a constant per fold, which must score chance — came out at ROC 0.473, and every ranking metric inherited the bias. Ranking each score within its own fold puts the control at exactly 0.500 and raises read-across from 0.803 to 0.828. Both numbers are published side by side and a regression test prevents the raw version returning. We also found and fixed a conformal procedure calibrated on a different model from the one making the predictions, and an operating threshold taken from the evaluation set's own labels.



Left: the spread a single 5-fold split cannot show; the median is the number to quote. Middle and right: two ways of asking whether the model generalises to unfamiliar chemistry rather than recognising close relatives. The structural forest ignores the similarity matrix, so it is flat in the middle panel by construction — an internal control. Ranks are pooled within fold, so a constant predictor scores exactly 0.500.

One split is not evidence. Repeating the whole evaluation over 20 fold assignments gives a median ROC of **0.776 (0.703–0.828)**. The pre-declared split landed at the optimistic end, so 0.776 is the number to quote for accuracy. The

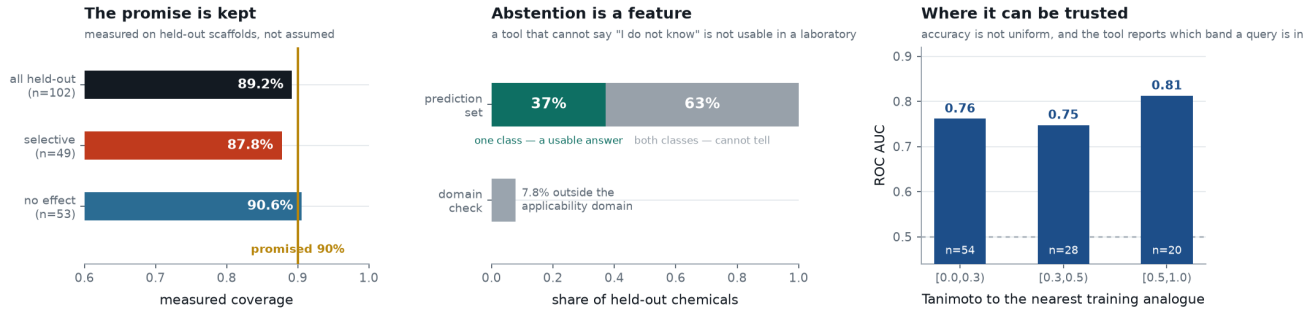
practical figure barely moves: **32 assays, range 31–37**, against 52 for every split.

Is it memorising close relatives? Read-across votes on measured analogues, so this is the obvious worry. Hiding every training analogue above a similarity cap answers it: at a cap of 0.7, ROC 0.820; at 0.5, ROC 0.799. There was little to memorise anyway — the median nearest-analogue Tanimoto is 0.36 and only 4% of chemicals have an analogue above 0.9. The structural forest ignores the similarity matrix and is flat across the cap: the internal control this test needs.

A harder split. 72% of Bemis-Murcko scaffold groups hold a single chemical, so scaffold-held-out is close to leave-one-out. Holding out whole Butina clusters at 0.6 similarity drops performance to 0.74–0.77 — still far above chance, but the three model families converge. **Read-across being the best model is specific to the scaffold split; structure carrying real signal is not.**

6 · Saying "I don't know"

A triage tool that answers every question is not calibrated, it is guessing. Coverage is **measured on held-out scaffolds, not assumed**: 89.2% against the 90% promised (87.8% and 90.6% by class). With 102 held-out chemicals the standard error on a coverage of 0.90 is about 3 points, so these agree. The honest cost: the prediction set is uninformative for 63% of chemicals and 7.8% fall outside the applicability domain. Those are the queries where the right answer is a shrug, and the tool gives one.



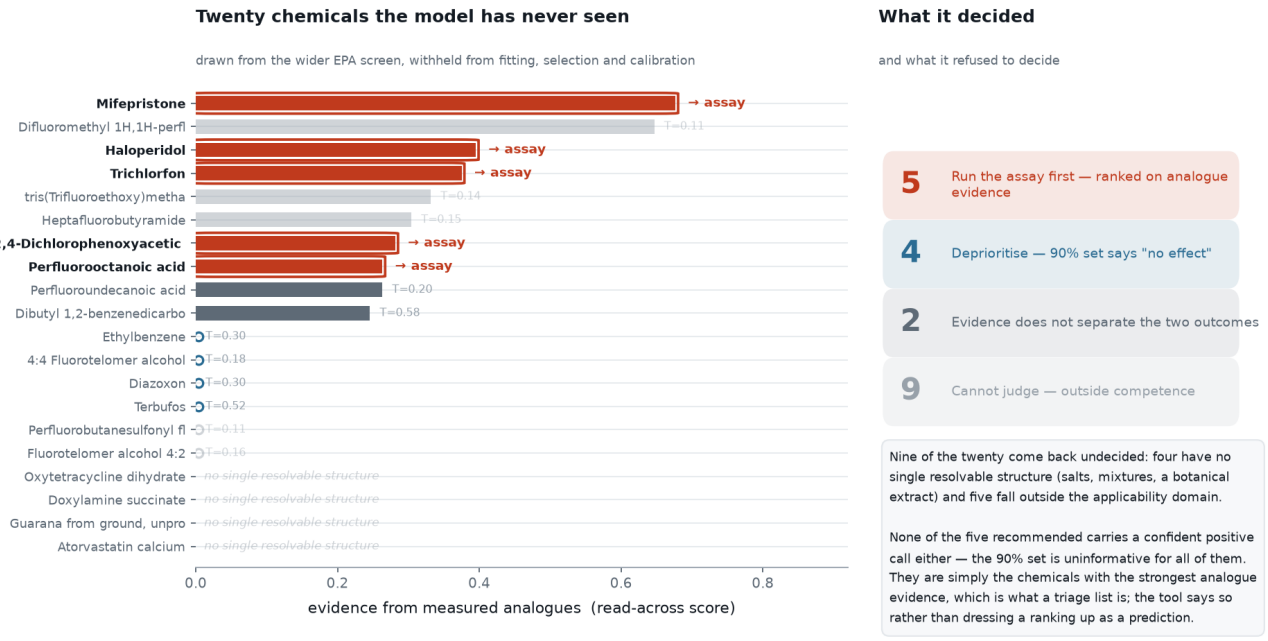
Mondrian (class-conditional) conformal prediction at a nominal 90% confidence, calibrated inside each training fold on the same fitted model that scores the held-out chemicals, and measured on held-out scaffolds. With 102 held-out chemicals the standard error on a coverage of 0.90 is about 3 points, so the measured value agrees with the promise.

7 · Scope, and what lies outside it

The 28 chemicals whose network effect appears only at lethal concentrations are excluded from the primary comparison, and they are not a random 28: they carry 22 scaffolds that then appear nowhere in the modelled set, and they form a coherent mechanistic family — rotenone, valinomycin, tributyltin, five strobilurins, two statins, chlordane, endosulfan, MPP^+. So the harder question is asked as well: **selective activity against everything else, cytotoxicants included.** Read-across scores **0.696 (0.595–0.784)** there, 44 assays against 65.5 at random — lower, as it should be, with the interval still excluding chance. The model does rank several of those cytotoxicants highly, and that is worth reading carefully: permethrin sits at a selectivity ratio of 1.9×, rotenone at 2.1×, just under the three-fold line. A high score is the model correctly recognising real neuroactive chemistry; what structure alone does not resolve is the fine selectivity margin. In the EPA protocol viability is read from the same plate, so such an assay still returns a usable answer.

8 · SciencePal in the core step

The central decision of this project does not happen in our repository — it happens inside SciencePal. The platform's **Biology Agent** is given the 102 measured analogues with their outcomes, 20 chemicals from the wider EPA screen withheld from every stage of fitting, selection and calibration, the project's scoring script and a task specification. Inside the SciencePal sandbox it executes that code, resolves each candidate structure, applies the model and the conformal rule, consults the literature on the nearest analogues, and returns a structured triage report: five chemicals to assay, with evidence, uncertainty, reason and the next assay to run. **Pre-declared ablation:** automatic ranking alone versus ranking + agent on the same 20 withheld chemicals, so the platform's contribution is measured rather than asserted.



The five recommended are the highest-evidence chemicals that pass the applicability-domain check; the ranking, the three nearest measured analogues behind each score and the 90% conformal set are written out for every candidate in results/agent_package/scores.csv, so the recommendation can be audited rather than trusted.

Nine of the twenty come back undecided — four have no single resolvable structure, five are outside the applicability domain — and none of the five recommended carries a confident positive call either. They are the chemicals with the strongest analogue evidence, which is what a triage list is, and the agent reports it that way rather than dressing a ranking up as a prediction.

9 · Limitations

102 chemicals. Every interval is wide; the per-fold standard deviation of ROC is 0.108. This is a small-data problem and is reported as one. **Cohort prevalence is 48%** because the EPA enriched it with suspected actives, so the assay saving does not transfer unchanged to an unselected library, and we do not claim it does. **Concordance, not external validation:** all 136 labelled chemicals sit inside the wider EPA screen of 460, so the second dataset is a cohort concordance check. **One assay, one laboratory, one developmental window. Not a safety verdict:** a negative in vitro result does not exclude developmental neurotoxicity — the OECD guidance says so directly.

10 · Next steps

(1) Extend to the 324 chemicals of the wider screen that have network data but no viability measurement — a weaker label, requiring an explicit statement of why, and a natural test of whether the ranking survives a noisier outcome. **(2)** Prospective validation: the five recommendations in §8 are a falsifiable prediction, and one MEA plate would settle them. **(3)** A pre-declared graph-neural-network comparison, deliberately left out at n = 102, once there is enough data to make it meaningful. **(4)** Cross-laboratory concordance between the NTP and ToxCast cohorts inside the EPA screen.

Originality and resources

All code, figures, results and text were created during the event. The author has a separate, unrelated project (NeuroChip Twin) entered in another competition; **no code, model, preprocessing, text, figure or video from it is used here** — DNT-Scout is a clean-room project with a new question, a new repository and new code. Data: US EPA network-formation assay (Shafer et al. 2019, DOI 10.23719/1503191) and PubChem, both public domain, re-downloaded from source and pinned by SHA256; raw data are not redistributed. Software: RDKit, scikit-learn, NumPy, pandas, Matplotlib — all open-source and permissively licensed. **No pretrained models, no proprietary models, no paid APIs.** Claude (Anthropic) was used as a programming and writing assistant under the author's direction; every scientific claim was checked against executed code, and every number in this document is produced by a script in the public repository. Where a result turned out weaker than first reported, the weaker number is the one published. Solo entry; full declaration in **docs/resource_declaration.md**.

Reproducibility: **pip install -e .** then four scripts regenerate every number and figure above; **pytest** runs 21 tests, four of which exist specifically to stop the defects described in §5 from coming back.